Numerical Analysis I

Quiz 12A Name:_____

**Question 1.**
Consider the following quadrature rule on $[a, b]$. Let

$$h = \frac{b-a}{4}, \qquad x_i = a + ih, \quad i = 0, 1, 2, 3, 4.$$

The quadrature rule is given by

$$\int_a^b f(x)\,dx \approx \frac{2h}{45}\left[7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)\right].$$

For $f \in C^6([a, b])$, the error is given by

$$E(f) = -\frac{8h^7}{945} f^{(6)}(\xi),$$

for some $\xi \in (a, b)$, where $E(f)$ denotes the exact integral minus the quadrature approximation.
This is called **Boole's rule**. This is the Closed Newton–Cotes rule with 5 nodes (4 subintervals).
Now let $n$ be a positive integer, and redefine

$$h = \frac{b-a}{4n}, \qquad x_i = a + ih, \quad i = 0, \ldots, 4n.$$

1. Construct the composite quadrature rule for $n = 2$. Express your answer as a weighted sum of $f(x_0), f(x_1), \ldots, f(x_8)$.
2. Generalize your result to an arbitrary positive integer $n$. Express the composite quadrature rule $Q_n(f)$ as a weighted sum of function values.
3. Using the error formula for the basic rule, show that the error of the composite rule satisfies

$$\left| \int_a^b f(x)\,dx - Q_n(f) \right| \le Ch^6,$$

where $C$ is independent of $n$. Determine $C$ explicitly in terms of $a$, $b$, and $\max_{x\in[a,b]} |f^{(6)}(x)|$.
4. If the step size $h$ is halved, by what factor does the error bound decrease? Justify your answer.

1. For $n = 2$, we have $h = (b-a)/8$. We apply the basic quadrature rule on $[x_0, x_4]$ and $[x_4, x_8]$:

$$\begin{aligned} Q_2(f) = {} & \frac{2h}{45}\left[7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)\right] \\ & + \frac{2h}{45}\left[7f(x_4) + 32f(x_5) + 12f(x_6) + 32f(x_7) + 7f(x_8)\right]. \end{aligned}$$

Combining the terms, we obtain

$$\boxed{\begin{aligned} Q_2(f) = \frac{2h}{45}\big[ & 7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) \\ & + 14f(x_4) + 32f(x_5) + 12f(x_6) \\ & + 32f(x_7) + 7f(x_8)\big]. \end{aligned}}$$

2. For an arbitrary positive integer $n$, we apply the basic quadrature rule on each interval $[x_{4j}, x_{4j+4}]$, for $j = 0, \ldots, n-1$.

Since each interval has length $4h$, we obtain

$$\boxed{Q_n(f) = \frac{2h}{45}\sum_{j=0}^{n-1}\left[7f(x_{4j}) + 32f(x_{4j+1}) + 12f(x_{4j+2}) + 32f(x_{4j+3}) + 7f(x_{4j+4})\right].}$$

Equivalently, combining the contributions at shared endpoints,

$$\boxed{\begin{aligned} Q_n(f) = \frac{2h}{45}\Bigg[&7f(x_0) + 7f(x_{4n}) \\ &+ 32\sum_{j=0}^{n-1}\left(f(x_{4j+1}) + f(x_{4j+3})\right) \\ &+ 12\sum_{j=0}^{n-1} f(x_{4j+2}) \\ &+ 14\sum_{j=1}^{n-1} f(x_{4j})\Bigg]. \end{aligned}}$$

3. On each interval $[x_{4j}, x_{4j+4}]$, the error formula for the basic rule gives

$$E_j = -\frac{8h^7}{945}f^{(6)}(\xi_j),$$

for some $\xi_j \in (x_{4j}, x_{4j+4})$.

The composite error is the sum of the local errors:

$$E_n(f) = \sum_{j=0}^{n-1} E_j.$$

Therefore,

$$\begin{aligned} |E_n(f)| &\le \sum_{j=0}^{n-1}|E_j| \\ &\le n\frac{8h^7}{945}\max_{x\in[a,b]}|f^{(6)}(x)| \\ &= \frac{4nh^7}{4}\frac{8}{945}\max_{x\in[a,b]}|f^{(6)}(x)| \\ &= \frac{2(b-a)}{945}h^6\max_{x\in[a,b]}|f^{(6)}(x)|. \end{aligned}$$

Thus,

$$\boxed{|E_n(f)| \le Ch^6, \qquad C = \frac{2(b-a)}{945}\max_{x\in[a,b]}|f^{(6)}(x)|.}$$

Since $C$ is independent of $n$, the composite quadrature rule is sixth-order accurate.

4. Since the error bound is proportional to $h^6$, halving $h$ gives

$$\left(\frac{h}{2}\right)^6 = \frac{h^6}{64}.$$

Therefore, the error bound decreases by a factor of 64.

**Question 2.**
Consider the following open quadrature rule on $[a, b]$. Let

$$h = \frac{b-a}{5}, \qquad x_i = a + ih, \quad i = 0, \ldots, 5.$$

The quadrature rule is given by

$$\int_a^b f(x)\,dx \approx \frac{5h}{24}\left[11f(x_1) + f(x_2) + f(x_3) + 11f(x_4)\right].$$

1. Determine the degree of exactness of this quadrature rule. Justify your answer.
2. What do you expect the order of accuracy of this quadrature rule to be in terms of $h$? No proof is required.

1. By a change of variables, we can work on $[0, 5]$ with nodes 1, 2, 3, and 4. The quadrature rule becomes

$$Q(p) = \frac{5}{24}\left[11p(1) + p(2) + p(3) + 11p(4)\right].$$

For $p(t) = 1$,

$$\int_0^5 1\,dt = 5 = Q(1).$$

For $p(t) = t$,

$$\int_0^5 t\,dt = \frac{25}{2} = Q(t).$$

For $p(t) = t^2$,

$$\int_0^5 t^2\,dt = \frac{125}{3} = Q(t^2).$$

For $p(t) = t^3$,

$$\int_0^5 t^3\,dt = \frac{625}{4} = Q(t^3).$$

However, for $p(t) = t^4$,

$$\int_0^5 t^4\,dt = 625,$$

while

$$\begin{aligned} Q(t^4) &= \frac{5}{24}\left[11 + 16 + 81 + 11(256)\right] \\ &= \frac{3655}{6}. \end{aligned}$$

Since these are not equal, the rule is not exact for polynomials of degree 4.

Therefore, the degree of exactness is

$$\boxed{d = 3.}$$

2. Since the rule is exact for polynomials of degree at most 3, the error in approximating the integrand is $O(h^4)$. Integration over an interval of length $5h$ contributes an additional factor of $h$. Therefore, the quadrature error is

$$\boxed{O(h^5).}$$